Lecture 4 (17/05/2021)

Permutation and Inverse Matrices

* Permutation matrix are used for exchanging rows.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \text{exchange rows} \longrightarrow \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

* we need to multiply the given matrix with another matrix to obtain matrix with exchanged rows.

$$\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

Doing the same operation on identity matrix,

$$\begin{bmatrix} \boxed{0 \quad 1} \\ \boxed{1 \quad 0} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \boxed{0 \quad 1} \\ \boxed{1 \quad 0} \end{bmatrix}$$

None of 1st row
Once of 2nd row

Once of 1st row
None of 2nd row

The same logic can be applied here,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

None of first row
Once of second row

Once of first row
None of second row

For a 3x3 matrix,

last row

same

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 9 & 10 & 11 \end{bmatrix} = \begin{bmatrix} = 9 & 10 & 11 \\ 4 & 5 & 6 \\ = 1 & 2 & 3 \end{bmatrix}$$

first row

Column Exchange

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \text{exchange columns} \rightarrow \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

→ Since it is a column operation it is to be operated to the Right side of the given matrix.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix} \begin{matrix} \text{None of 1st col}^n \\ \text{Once of 2nd col}^n \\ \text{Once of 1st col}^n \\ \text{None of 2nd col}^n \end{matrix}$$

Shortcut:

If we exchange the desired row/column of it's corresponding identity matrix and multiply with it, we will obtain the matrix with exchanged row/column. (replicate operation on identity matrix) [Works on 3x3 matrix too]

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

col^n operation

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

row operation

Inverse Matrices

From previous lecture, E_{21} was the matrix which is responsible for making the 2,1 th element of matrix 0.

In order to undo the operation done by E_{21}, we can multiply the product matrix with another matrix.

What would that matrix be?

$$\square \times E_{21} = I$$

Identity matrix
- No significant operation done
- Neutral operation

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I}$$

If matrix A does linear operation on any space and another matrix B that undoes the operation of A, then B is called the inverse matrix of A. $\boxed{B = A^{-1}}$ (Concept of inverse.)

Inverse Matrix

- Only for square matrices
- Not all matrices are invertible (must be non-singular)
- If there's a matrix A that satisfies the above conditions, then there must exist a matrix A^{-1} such that,

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

* undoes the activity of each other.

If a matrix has an inverse, the matrix is a non-singular / invertible matrix.

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Non-invertible. Why?

$\det(A) = 1.6 - 2.3 = 0$ (singular matrix)

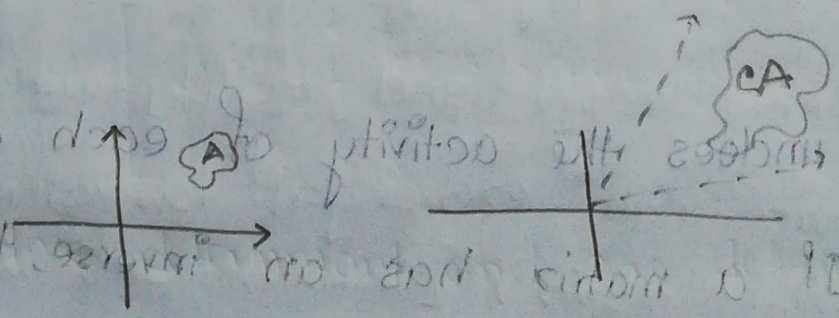
Determinant

From the perspective of linear transformation, when a matrix does linear transformation, it stretches or squeezes the space.

Finding the factor by which the previous area increase or decrease

Example:

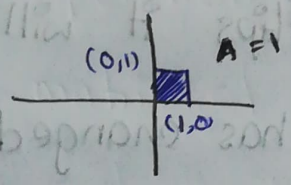
$$\begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$$



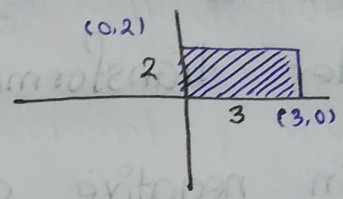
transformed (stretched by c)

Example:

$$\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$



Before Transformation



$cA = 6$ (2×3)

After Transformation

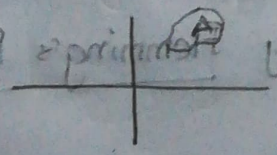
$$\boxed{c = 6}$$

- The special scaling factor (c) by which a linear transformation changes any area is called the <u>determinant of the transformation</u>.

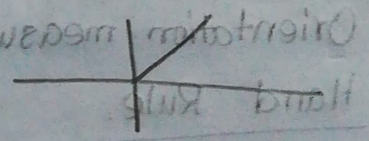
  if det > 1, area <u>stretched</u> after transformation

  det < 1, area <u>squeezed</u> after transformation

- If the determinant of any transformation is 0, it means it <u>squeezes</u> all the space into <u>a line</u> (since line occupies no area)



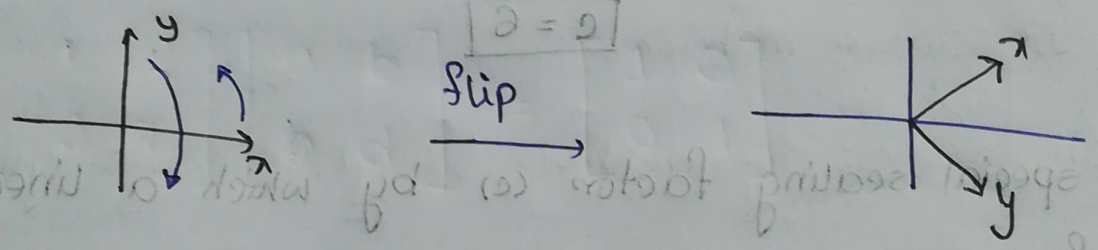
$\rightarrow$



$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

- Determinant also allows negative value. It has to do with the idea of orientation.

If after transformation, the orientation flips it will result in negative determinant since it has changed the orientation of every space in the 3D space



- What about 3 x 3 matrices?

for 2 x 2 Matrices, the area was stretched / squeezed

for 3 x 3 Matrices, the volume is stretched / squeezed.

(gets scaled)

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- Same idea of negative determinant and zero determinant goes for 3D space as well.

Negative determinant - Orientation measured by Flemming's Right Hand Rule.

- What linear transformation is done by the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ in perspective of determinants?

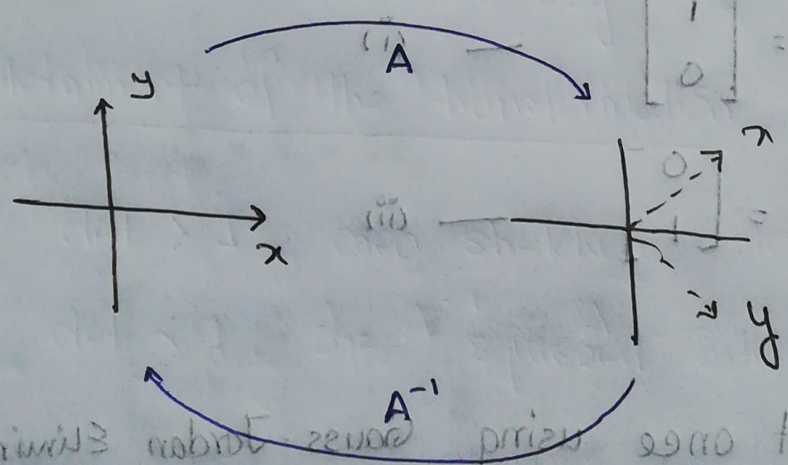
  - Squeezes the area into a line.

Inverses

$A \rightarrow$ does linear transformation

$A^{-1} \rightarrow$ does the opposite linear transformation

(takes the space back to it's previous form)



Why matrix with $\det(A) = 0$ can't have inverse?

- As the matrix with 0 determinant squeezes the space into a line, the inverse can't take it back to it's previous state (shape deformed)

$\rightarrow$ can't make anything out of 0.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \qquad \det(A) = 1.7 - 2.3 = 1 \neq 0$$

$\therefore A^{-1}$ exists.

$$A \cdot A^{-1} = I \qquad [\text{col}^n \text{ operation}]$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{— (i)}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{— (ii)}$$

Solving both eq^n at once using Gauss-Jordan Elimination

$$\left[\begin{array}{cc:cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array}\right]$$ Augmented matrix

$R_2 = R_2 - 3.R_1$

$$\left[\begin{array}{cc:cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array}\right]$$

For A^{-1} we have to do upward elimination. to get I at left side.

$R_1 = R_1 - 2R_2$

$$\left[\begin{array}{cc:cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array}\right]$$

A^{-1}

Started with $[A : I] = [I : A^{-1}]$ Ended with

$$\therefore \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} = A^{-1}$$